

**Mass extinction OTW**

A geophysics professor from MIT claims we're on our way to a mass extinction event.
<http://dgit.in/WeDinoDead>

**T2 = T6?**

Linda Hamilton and Arnold Schwarzenegger returning for the new Terminator movie set to release in 2019. <http://dgit.in/T2BRBAgain>

All charged up

The humble battery can be traced back to some very interesting discoveries. This is a tale of how dead frogs helped us make batteries!

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erhaps one of the most recurring narratives in today's discussion about the planet is how much

we've screwed up when it comes to human-caused climate change. We're all aware that fossil fuels are partly to blame for that. So what's stopping the sustainable energy revolution? Why aren't we living off all green energy already? Are you fed up of the rhetorical questions? We'll tell you where the major push to get clean energy into our vehicles and homes really gets stuck... the humble battery!

From your smartphone to the international space station, batteries are at the heart of pretty much all modern technology. While we have covered battery tech in many articles over the years, and are always hoping the next big thing in batteries will change everything... this article is not about all that. Here we're going to look back at the origin story of one of the most important inventions of the modern world, and we hope the story is electrifying in more ways than the obvious!

The basic working principle of the battery hasn't really changed

at all. A battery still consists mainly of three components – a negative electrode (anode), a positive electrode (cathode) and a conducting material (electrolyte). The electricity is generated as electrons move from the anode to the cathode because of the chemical reactions between the materials used.

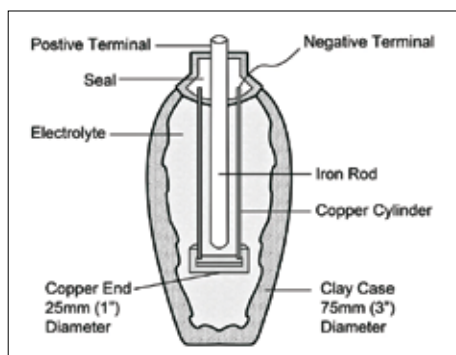
PRE-HISTORY

While no ancient pencil cells have been discovered, the use of electric charge might be a lot older than we are led to believe. For example, in a 1936 building project for a railroad in Baghdad, an undiscovered tomb was unearthed. Inside was found what's now known as the Parthian battery – a clay jar, sealed with pitch at the top, with an

iron rod sticking out the centre which was encased in a cylindrical tube of wrapped copper sheet. It was just 15 cm tall, and the copper tube was about 4 cm in diameter by 12 cm in length. When replicas were filled with an acidic liquid such as vinegar, the apparatus was able to produce between 1.5 and 2 volts. The design is not too different from the Leyden jars of the mid-1700s. Although it's thought that this primitive battery could have been used to electroplate gold onto silver artifacts, some scientists believe that's just a mistaken belief and suggest that it was used to store sacred scrolls. Even if it was used for electrolysis, it's still not really a battery, per se.

AN AMPHIBIAN TOUCH

A basic electric phenomenon that would form the basis of battery technology for a long time ended up being discovered inadvertently – and labelled incorrectly at that. In 1786 Luigi Galvani, an Italian physicist, while dissecting a frog attached to a brass hook with an iron scalpel noticed its leg twitch whenever he touched it. He labelled this as animal electricity, crediting the leg itself as the source. Alessandro Volta was quick to correct him with the concept that this was caused due to the two metals involved and a conducting liquid medium in between. He even went ahead and car-



Parthian battery explained

Source: <http://www.arri.org/images>